

PROJECT REPORT OF CO-OP Project at Industry (Module-2)
ON
NODWISE: A Smart ML Framework for Continuous Head Movement Awareness
submitted in partial fulfilment of the requirements for the award of degree of
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In
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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **“NODWISE: A Smart ML Framework for Continuous Head Movement Awareness”** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Swati Goel. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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Abstract :

The issue of cervical pain is rapidly becoming an increasing concern among computer and digital technology users in the workplace. Screen time, limited neck rotation, and poor posture are the cause of what is commonly referred to as tech neck. The existing ergonomic tools or posture-tracking devices are mostly passive, require any additional wearables, or do not provide timely neck-centered feedback.

The proposed project presents Nodwise, a camera-based system that operates on machine learning to monitor the motions of the neck and the instances when a user remains in a still motion, which is too long. Nodwise constantly monitors the neck position and movement using a pose-estimation model after user login the camera setup.

When the neck remains in a certain position of above a given threshold value, then the system sounds an alarm or displays a visual warning, and recommends some easy neck exercises. It remains silent in case of good posture. In a small pilot study, the Nodwise was demonstrated to be able to recognize long segments of bad neck posture and give timely feedback, and the users of the device claimed they experienced more awareness of their posture, and they had improved neck mobility.

The project demonstrates that a machine-learning system, based on low-cost, non-contact, and utilizing machine-learning technology can be used to prevent neck issues in current computer-based employment.

1. Introduction :

1.1 Background of the Study :

With the rapid increase in digital device usage, people spend long hours in front of computers, laptops, and smartphones for work, education, and entertainment. This has led to a rise in neck strain, cervical pain, and posture-related problems, commonly known as “tech neck.” Poor sitting posture, forward head tilt, and prolonged immobility are major causes of these issues.

Continuous screen exposure places stress on the cervical spine and neck muscles, leading to discomfort, stiffness, and long-term musculoskeletal problems. Traditional ergonomic solutions often rely on passive awareness and lack real-time monitoring. Therefore, there is a growing need for smart, technology-driven systems that can actively monitor posture, detect unhealthy neck movement patterns, and promote better ergonomic habits.

1.2 Problem Statement :

Current posture correction methods such as ergonomic chairs, posture braces, or wearable devices are often expensive, passive, or inconvenient for continuous daily use. Many of these solutions depend heavily on user awareness and discipline, making them less effective for long-term posture improvement. Wearable devices may also cause discomfort, require maintenance, or fail to provide natural, non-intrusive monitoring.

As a result, there is a growing need for a low-cost, camera-based intelligent system capable of actively monitoring neck movement, detecting prolonged poor posture or inactivity, and providing real-time corrective interventions. Such a system can offer accessible, non-invasive, and practical support for preventing cervical pain and promoting healthier posture habits.

1.3 Objectives of the Study :

The primary objectives are:

- To develop a machine learning-based neck posture monitoring system
- To detect prolonged static neck posture in real-time
- To alert users about poor posture or inactivity
- To suggest simple neck exercises for correction
- To reduce the long-term risk of cervical pain and musculoskeletal disorders

1.4 Scope of the Project :

This project focuses on:

- Real-time head and neck movement detection
- Webcam-based posture analysis
- Alert systems for posture correction
- User ergonomic awareness

It excludes:

- Autonomous farming systems
- Real-time drone or IoT sensor integration
- Large-scale agricultural monitoring systems

2. Methodology :

2.1 System Overview :

The proposed Nodwise system follows these stages:

1. User Authentication
2. Camera Calibration
3. Continuous Head Movement Tracking
4. Neck Angle and Mobility Analysis
5. Inactivity Detection
6. Alert Generation
7. Corrective Exercise Suggestion

2.2 Pose Detection

The system uses machine learning-based pose estimation to detect facial and neck landmarks through webcam input. Head orientation and movement patterns are tracked continuously. By analyzing key points such as head tilt, neck angle, and directional movement, the system can accurately assess posture in real time. This continuous tracking helps identify unhealthy neck positions, prolonged static posture, or reduced mobility. The approach is non-invasive, requiring only a standard camera, making it practical and accessible for everyday use.

2.3 Movement Analysis

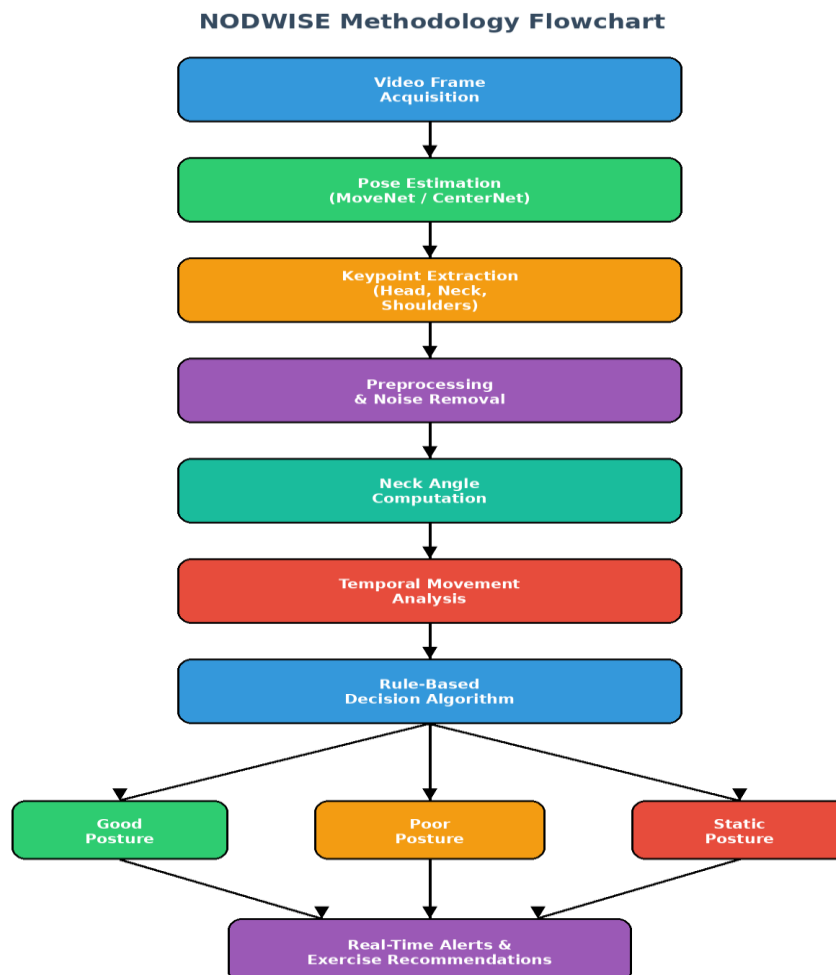
A predefined movement threshold and timer are used to evaluate whether sufficient neck motion is occurring. If neck movement remains below the threshold for extended periods, the system flags potential posture risk. This helps in identifying prolonged inactivity or poor ergonomic habits before they develop into serious cervical issues. By continuously comparing movement patterns against healthy posture standards, the system ensures timely detection and preventive intervention..

2.4 Alert Mechanism

- Visual notifications
- Audio alerts
- Exercise recommendations
- Posture correction reminders
- Real-time inactivity warnings
- Guided neck mobility suggestions

2.5 Non-Invasive Monitoring

When movement patterns remain healthy, no alerts are issued, ensuring minimal user interruption. This allows users to work comfortably without unnecessary distractions while maintaining natural productivity. The system operates silently in the background, intervening only when unhealthy posture or prolonged inactivity is detected. Such an approach makes the solution user-friendly, efficient, and suitable for long-term daily use.



3. Tools and Technologies :

The proposed NODWISE framework integrates machine learning, computer vision, and real-time video processing technologies to continuously monitor neck posture and movement. The system was designed using lightweight and efficient software tools that enable accurate posture analysis without requiring expensive hardware or wearable devices. The combination of artificial intelligence techniques and webcam-based monitoring allows the framework to operate as a low-cost, scalable, and user-friendly ergonomic solution.

3.1 Programming Environment

Python was selected as the primary programming language for implementing the proposed system because of its simplicity, flexibility, and extensive support for machine learning and computer vision applications. Python provides a rich ecosystem of libraries that simplify real-time image processing, pose estimation, mathematical computation, and alert generation.

The language also enabled rapid development and seamless integration between different modules of the framework, including posture tracking, temporal analysis, and notification systems. Its compatibility with deep learning frameworks and image-processing libraries made it highly suitable for building the NODWISE posture monitoring system.

3.2 Computer Vision and Image Processing

Computer vision forms the core of the proposed framework. The system continuously captures video frames using a webcam and analyzes them in real time to identify head and neck posture.

OpenCV was used for webcam access, video frame acquisition, image preprocessing, and frame handling operations. It enabled smooth real-time video processing and helped the system efficiently manage continuous monitoring tasks. OpenCV also supported image enhancement and frame consistency operations required for accurate pose estimation.

The use of computer vision eliminated the need for physical sensors and enabled completely non-invasive posture monitoring.

3.3 Pose Estimation Technologies

The proposed framework uses deep learning-based pose estimation models to detect important body landmarks such as the head, neck, and shoulders. Technologies such as MediaPipe, MoveNet, and BlazePose were used for extracting body keypoints from webcam frames in real time.

These models are based on Convolutional Neural Networks (CNNs), which are capable of accurately identifying spatial body positions even under varying lighting conditions and slight camera disturbances. The extracted landmarks are used for neck angle computation, movement tracking, and posture classification.

Pose estimation technology plays a major role in enabling the system to continuously monitor posture without requiring additional wearable devices or specialized hardware.

3.4 Machine Learning Techniques

The NODWISE framework integrates multiple machine learning and movement analysis techniques for intelligent posture evaluation.

Real-time pose estimation was used to continuously detect body landmarks and estimate neck orientation. The system also implemented temporal movement analysis, which monitors user movement patterns over time instead of evaluating posture using only single-frame analysis. This approach improves posture detection reliability and helps identify prolonged static posture more accurately.

A rule-based decision algorithm was developed to classify posture into three categories:

- Good Posture
- Poor Posture
- Static Posture

The classification process was based on neck angle thresholds, movement variance, and posture duration. This combination of spatial and temporal analysis significantly improved the accuracy and reliability of posture monitoring.

3.5 Mathematical and Numerical Processing

NumPy was used for performing mathematical computations and coordinate-based calculations within the framework. The library was utilized for:

- Vector calculations
- Neck angle computation
- Coordinate normalization
- Movement displacement analysis
- Motion variance calculation

These calculations were essential for determining posture orientation and evaluating movement patterns over time.

3.6 Temporal Analysis and Monitoring

The Time module was used to track inactivity duration, monitor posture timing, and perform temporal movement analysis. The framework continuously observed whether the user remained in the same neck position for extended periods.

By measuring movement variance across multiple frames, the system identified prolonged static posture conditions and generated corrective alerts whenever required.

This temporal monitoring mechanism helped the framework move beyond traditional frame-by-frame posture detection and enabled more intelligent ergonomic analysis.

3.7 Alert and Notification System

The system integrates a real-time feedback mechanism designed to improve ergonomic awareness and encourage posture correction.

Whenever unhealthy posture or prolonged inactivity is detected, the framework generates:

- Visual warning notifications
- Audio alerts
- Exercise recommendations
- Neck mobility reminders
- Posture correction prompts

The notification system was designed to remain non-intrusive and operate silently during healthy posture conditions to avoid unnecessary interruptions.

3.8 Hardware Requirements

One of the major advantages of the proposed NODWISE framework is its low hardware dependency. The system was designed to operate efficiently on standard consumer-grade hardware without requiring wearable devices or specialized posture-monitoring equipment.

The minimum hardware requirements include:

- Standard webcam for video input
- Laptop/Desktop computer for processing
- Audio output device for alerts and notifications

The framework maintained moderate CPU utilization during continuous monitoring, making it suitable for long-duration deployment in workplaces, educational environments, and home-office setups.

3.9 Advantages of the Selected Technologies

The technologies selected for the proposed system offer several advantages:

- Low-cost implementation
- Real-time monitoring capability
- Non-invasive posture tracking
- High posture detection accuracy
- Easy deployment and scalability
- Minimal hardware requirements
- Efficient machine learning integration
- Improved ergonomic awareness

4. Implementation :

4.1 User Authentication

Users log in before using the system. This ensures personalized access and allows the system to maintain individual session records for posture monitoring. Authentication also helps in storing user-specific preferences, activity duration, and performance history for better tracking and analysis.

4.2 Camera Calibration

The webcam initializes and aligns user posture for baseline tracking. During this stage, the system detects the user's face and neck landmarks to establish a reference posture position. Proper calibration improves detection accuracy and ensures reliable monitoring under different user positions and lighting conditions.

4.3 Continuous Monitoring

The system continuously observes neck angle, motion frequency, and posture duration. Real-time data from the webcam is processed to identify movement patterns, prolonged inactivity, or unhealthy neck positions. This ongoing analysis allows the system to provide timely alerts while ensuring smooth and uninterrupted posture supervision.

4.4 Alert Triggering

When insufficient movement is detected:

- Warning message displayed
- Audio notification played
- Exercise prompt shown
- Real-time posture correction reminder provided
- Neck mobility suggestions displayed on screen
- Prolonged inactivity alerts activated after threshold breach

These alerts are designed to immediately capture user attention and encourage corrective action before discomfort increases. The system ensures that notifications are timely yet non-intrusive, promoting healthier posture habits without significantly disrupting workflow.

4.5 System Workflow

1. User starts session
2. Camera captures and calibrates posture
3. ML model analyzes neck movement
4. System monitors posture continuously
5. Threshold exceeded → Alert generated
6. Warning and exercise prompt displayed
7. User corrects posture and monitoring resumes

5. Results

5.1 Experimental Setup

The proposed NODWISE framework was implemented using Python, OpenCV, and deep learning-based pose estimation models integrated with TensorFlow/PyTorch. The system was tested under real-time conditions using a standard webcam in normal indoor environments.

A pilot study was conducted with multiple users performing computer-based activities for approximately 1–3 hours per session. During testing, the system continuously monitored neck posture, movement frequency, static posture duration, and user mobility patterns while generating alerts whenever unhealthy posture or prolonged inactivity was detected.

5.2 Evaluation Metrics

The system was evaluated using several standard performance metrics including accuracy, precision, recall, F1-score, response time, frame rate (FPS), static posture detection rate, and false alert rate.

These metrics were used to measure the reliability, sensitivity, efficiency, and real-time performance of the proposed framework.

5.3 Core Performance Results

The proposed framework demonstrated strong performance across all major modules involved in posture monitoring and movement analysis.

The pose estimation model achieved the highest accuracy of 96.5%, showing highly reliable body landmark detection. Neck angle detection achieved an accuracy of 94.2%, while temporal movement analysis achieved 92.8% accuracy in identifying prolonged static posture.

The rule-based decision algorithm achieved an overall accuracy of 95.6%, successfully classifying posture into good posture, poor posture, and static posture categories. The static posture detection module also achieved 94.9% accuracy with low false alert rates.

Overall, all major components achieved performance above 92%, confirming the robustness and reliability of the system.

5.4 Time and Efficiency Analysis

The framework demonstrated efficient real-time processing capability during continuous monitoring.

Frame preprocessing achieved the lowest response time of approximately 0.7 seconds, while pose estimation required approximately 0.8 seconds. Temporal movement analysis required slightly higher processing time of around 1.5 seconds because of continuous movement tracking over time.

Despite this, all modules operated within acceptable real-time limits, making the system suitable for long-duration deployment.

5.5 Frame Rate Performance

The proposed NODWISE framework achieved an average processing speed of approximately 15–25 FPS, enabling smooth real-time posture monitoring while maintaining computational efficiency on standard hardware.

5.6 Static Posture Detection Analysis

The static posture detection module achieved a correct detection rate of 93.5% with a low false alert rate of approximately 5.2%. The system successfully differentiated between active posture conditions and prolonged static posture while minimizing unnecessary notifications.

These results demonstrate the effectiveness of temporal movement analysis in improving posture monitoring reliability.

5.7 User Behavior Improvement

The pilot study showed significant improvements in user posture behavior after using the system.

The average static posture duration reduced from 18.5 minutes to 10.2 minutes, while posture awareness scores improved from 5.8 to 8.3. Neck discomfort levels also reduced noticeably, and movement frequency increased significantly during prolonged computer usage.

These findings confirm that real-time monitoring and corrective feedback can effectively improve ergonomic habits and reduce cervical discomfort.

5.8 Overall Findings

The overall evaluation confirms that the proposed NODWISE framework provides an accurate, low-cost, and reliable solution for real-time neck posture monitoring and ergonomic awareness.

The integration of machine learning, computer vision, pose estimation, and temporal movement analysis enabled the framework to achieve strong performance while maintaining low hardware requirements and efficient real-time operation.

Another important technical finding was the framework's scalability and accessibility. The system operated efficiently on standard consumer-grade hardware using a webcam and Python-based libraries, without requiring wearable devices or specialized sensors. CPU utilization remained moderate during continuous monitoring, making it suitable for long-duration deployment in educational, workplace, and home environments.

6. Conclusion and Future Scope :

6.1 Conclusion

NODWISE successfully demonstrates the application of machine learning in real-time cervical pain prevention. By combining posture detection, continuous monitoring, and corrective alerts, the system offers an efficient, affordable, and accessible solution for modern digital lifestyles. The integration of computer vision and pose estimation enables accurate detection of unhealthy neck posture and prolonged inactivity without the need for expensive wearable devices or specialized hardware.

The system not only improves ergonomic awareness but also encourages healthier posture habits through timely visual, audio, and exercise-based interventions. Its non-invasive design ensures minimal disruption while providing continuous supervision, making it practical for students, office workers, and individuals with prolonged screen exposure.

Overall, NODWISE highlights the potential of AI-driven healthcare support systems in addressing everyday musculoskeletal challenges. By promoting preventive care rather than reactive treatment, the framework contributes toward reducing tech-neck risks, enhancing user well-being, and supporting healthier work environments in an increasingly technology-dependent world.

6.2 Future Scope

- Mobile app integration
- AI-based personalized posture recommendations
- Wearable device compatibility
- Cloud-based posture analytics
- Workplace productivity integration

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